

EC-390: Digital Signal Processing

LECTURE NO 11: Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT)

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Organization of Lecture

1. Introduction.
2. Discrete Fourier Transform (DFT) of Discrete Time Signal.
3. Properties of DFT.
4. Relation Between DFT and $\mathbf{Z}$ -Transform.
5. Analysis of LTI Discrete Time Systems Using DFT.
6. Fast Fourier Transform (FFT).
7. Decimation in Time (DIT) Radix-2 FFT.
8. Computation of Inverse DFT Using FFT.

1. Introduction

The discrete time Fourier transform (DTFT) discussed in lecture-10, provides a method to represent a discrete time signal in frequency domain and to perform frequency analysis of discrete time signal.

The drawback in DTFT is that the frequency domain representation of a discrete time signal obtained using DTFT will be a continuous function of ω and so it cannot be processed by digital system. The discrete Fourier transform (DFT) has been developed to convert a continuous function of ω to a discrete function of ω , so that frequency analysis of discrete time signals can be performed on a digital system.

Basically, the DFT of a discrete time signal is obtained by sampling the DTFT of the signal at uniform frequency intervals.

The samples of DTFT are represented as a function of integer k , and so the DFT is a sequence of complex numbers represented as $X(k)$ for $k = 0, 1, 2, 3, \text{etc.,}$.

1. Introduction

Since $X(k)$ is a sequence consisting complex numbers, the magnitude and phase of each sample can be computed and listed as magnitude sequence and phase sequence respectively. The graphical plots of magnitude and phase as a function of k are also drawn.

The plot of magnitude versus k is called *magnitude spectrum* and the plot of phase versus k is called *phase spectrum*. In general these plots are called *frequency spectrum*.

The drawback in *DFT* is that, the computation of each sample of DFT involves a large number of calculations and when large number of samples are required, the number of calculations will further increase. In order to overcome this drawback, a number of methods or algorithms have been developed to reduce the number of calculations. The various methods developed to compute DFT with reduced number of calculations are collectively called *Fast Fourier Transform* (FFT).

2. Discrete Fourier Transform (DFT) of Discrete Time Signal

2.1 Development of DFT from DTFT

The frequency domain representation of a discrete time signal obtained using discrete time Fourier transform (DTFT) will be a continuous and periodic function of ω , with periodicity of 2π .

The sampling frequency points are denoted as ω_k and they are given by:

$$\omega_k = \frac{2\pi k}{N} ; \quad \text{for } k = 0, 1, 2, \dots, N-1$$

Now, the DFT is a sequence consisting of N-samples of DTFT. Let the samples are denoted by $X(k)$ for $k = 0, 1, 2, \dots, N-1$. Therefore, the sampling of $X(e^{j\omega})$ is mathematically expressed as,

$$X(k) = X(e^{j\omega}) \Big|_{\omega = \frac{2\pi k}{N}} ; \quad \text{for } k = 0, 1, 2, \dots, N-1 \dots (2.1)$$

The DFT sequence starts at $k = 0$, corresponding to $\omega = 0$. Generally, the DFT is defined along with number of samples and is called ***N-point DFT***. The number of samples N for a finite duration sequence $x(n)$ of length L should be such that, $N \geq L$, in order to avoid aliasing of frequency spectrum.

2. Discrete Fourier Transform (DFT) of Discrete Time Signal

2.2 Definition of Discrete Fourier Transform (DFT)

Let, $x(n)$ = Discrete time signal of length L

$$X(k) = \text{DFT of } x(n)$$

Now, the *N-point DFT* of $x(n)$, where $N \geq L$, is defined as,

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi kn}{N}} ; \quad \text{for } k = 0, 1, 2, \dots, N-1 \quad \dots(2.2)$$

Symbolically the N -point DFT of $x(n)$ can be expressed as,

$$\mathcal{DFT}\{x(n)\}$$

where, $\mathcal{DFT}$ is the operator that represents discrete Fourier transform.

$$\therefore \mathcal{DFT}\{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi kn}{N}} ; \quad \text{for } k = 0, 1, 2, \dots, N-1$$

Since $X(k)$ is a sequence consisting of N -complex numbers for $k = 0, 1, 2, \dots, N-1$, the DFT of $x(n)$ can be expressed as a sequence as shown below.

$$X(k) = \{X(0), X(1), X(2), \dots, X(N-1)\}$$

2. Discrete Fourier Transform (DFT) of Discrete Time Signal

2.3 Frequency Spectrum Using DFT

The $X(k)$ is a discrete function of discrete time frequency ω , and so it is also called discrete frequency spectrum (or signal spectrum) of the discrete time signal $x(n)$.

The $X(k)$ is a complex valued function of k and so it can be expressed in rectangular form as,

$$X(k) = X_r(k) + jX_i(k)$$

where, $X_r(k)$ = Real part of $X(k)$

$X_i(k)$ = Imaginary part of $X(k)$

Now the Magnitude function (or Magnitude spectrum) $|X(k)|$ is defined as,

$$|X(k)|^2 = X(k) X^*(k) \quad \text{or} \quad |X(k)|^2 = X_r^2(k) + X_i^2(k)$$

$$\therefore |X(k)| = \sqrt{X(k) X^*(k)} \quad \text{or} \quad \sqrt{X_r^2(k) + X_i^2(k)}$$

2. Discrete Fourier Transform (DFT) of Discrete Time Signal

The Phase function (or Phase spectrum) $\angle X(k)$ is defined as,

$$\angle X(k) = \text{Arg}[X(k)] = \tan^{-1} \left[\frac{X_i(k)}{X_r(k)} \right]$$

Since $X(k)$ is a sequence consisting of N -complex numbers for $k = 0, 1, 2, \dots, N-1$, the magnitude and phase spectrum of $X(k)$ can be expressed as a sequence as shown below.

Magnitude sequence, $|X(k)| = \{|X(0)|, |X(1)|, |X(2)|, \dots, |X(N-1)|\}$

Phase sequence, $\angle X(k) = \{\angle X(0), \angle X(1), \angle X(2), \dots, \angle X(N-1)\}$

The magnitude and phase sequence can be sketched graphically as a function of k .

The plot of samples of magnitude sequence versus k is called ***magnitude spectrum*** and the plot of samples of phase sequence versus k is called ***phase spectrum***. In general these plots are called ***frequency spectrum***.

2. Discrete Fourier Transform (DFT) of Discrete Time Signal

2.4 Inverse DFT

Let, $x(n)$ = Discrete time signal

$X(k)$ = N-point DFT of $x(n)$

The inverse discrete Fourier transform of the sequence $X(k)$ of length N is defined as,

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi kn}{N}} ; \text{ for } n = 0, 1, \dots, N-1 \quad \text{.....(2.3)}$$

3. Properties of DFT

1. Linearity

Let, $\mathcal{DFT}\{x_1(n)\} = X_1(k)$ and $\mathcal{DFT}\{x_2(n)\} = X_2(k)$ then by linearity property,
 $\mathcal{DFT}\{a_1 x_1(n) + a_2 x_2(n)\} = a_1 X_1(k) + a_2 X_2(k)$, where a_1 and a_2 are constants.

2. Periodicity

Hence, if $x(n)$ and $X(k)$ are N point DFT pair then,

$$x(n + N) = x(n) \quad ; \quad \text{for all } n$$

$$X(k + N) = X(k) \quad ; \quad \text{for all } k$$

3. Circular time shift

$$\text{if } \mathcal{DFT}\{x(n)\} = X(k), \text{ then } \mathcal{DFT}\{x((n - m))_N\} = X(k) e^{-j \frac{2\pi km}{N}}$$

3. Properties of DFT

4. Time reversal

if $\mathcal{DFT}\{x(n)\} = X(k)$, then $\mathcal{DFT}\{x(N-n)\} = X(N-k)$.

5. Conjugation

if, $\mathcal{DFT}\{x(n)\} = X(k)$, then $\mathcal{DFT}\{x^*(n)\} = X^*(N-k)$.

6. Circular frequency shift

if $\mathcal{DFT}\{x(n)\} = X(k)$ then $\mathcal{DFT}\left\{x(n) e^{j\frac{2\pi mn}{N}}\right\} = X((k-m))_N$

7. Circular convolution

Let $\mathcal{DFT}\{x_1(n)\} = X_1(k)$ and $\mathcal{DFT}\{x_2(n)\} = X_2(k)$, then by convolution property,

$$\mathcal{DFT}\{x_1(n) \circledast x_2(n)\} = X_1(k) X_2(k)$$

3. Properties of DFT

Table 1 : Properties of Discrete Fourier Transform (DFT)

Note : $X(k) = \mathcal{DFT}\{x(n)\}$; $X_1(k) = \mathcal{DFT}\{x_1(n)\}$; $X_2(k) = \mathcal{DFT}\{x_2(n)\}$; $Y(k) = \mathcal{DFT}\{y(n)\}$

Property	Discrete time signal	Discrete Fourier Transform
Linearity	$a_1 x_1(n) + a_2 x_2(n)$	$a_1 X_1(k) + a_2 X_2(k)$
Periodicity	$x(n + N) = x(n)$	$X(k + N) = X(k)$
Circular time shift	$x((n - m))_N$	$X(k) e^{\frac{-j2\pi k m}{N}}$
Time reversal	$x(N - n)$	$X(N - k)$
Conjugation	$x^*(n)$	$X^*(N - k)$
Circular frequency shift	$x(n) e^{\frac{j2\pi m n}{N}}$	$X((k - m))_N$
Multiplication	$x_1(n) x_2(n)$	$\frac{1}{N} [X_1(k) \otimes X_2(k)]$

3. Properties of DFT

Circular convolution	$x_1(n) \circledast x_2(n) = \sum_{m=0}^{N-1} x_1(m) x_2((n-m))_N$	$X_1(k) X_2(k)$
Circular correlation	$\bar{r}_{xy}(m) = \sum_{n=0}^{N-1} x(n) y^*((n-m))_N$	$X(k) Y^*(k)$
Symmetry of real signals	$x(n) \text{ is real}$	$X(k) = X^*(N-k)$ $X_r(k) = X_r(N-k)$ $X_i(k) = -X_i(N-k)$ $ X(k) = X(N-k) $ $\angle X(k) = -\angle X(N-k)$
Symmetry of real and even signal	$x(n) \text{ is real and even}$ $x(n) = x(N-n)$	$X(k) = X_r(k) \text{ and } X_i(k) = 0$
Symmetry of real and odd signal	$x(n) \text{ is real and odd}$ $x(n) = -x(N-n)$	$X(k) = jX_i(k) \text{ and } X_r(k) = 0$
Parseval's relation	$\sum_{n=0}^{N-1} x_1(n) x_2^*(n)$	$\frac{1}{N} \sum_{k=0}^{N-1} X_1(k) X_2^*(k)$

4. Relation Between DFT and Z-Transform

The Z-transform of N-point sequence $x(n)$ is given by,

$$\mathcal{Z}\{x(n)\} = X(z) = \sum_{n=0}^{N-1} x(n)z^{-n}$$

$$z = e^{j\omega_k}; \text{ where } \omega_k = \frac{2\pi k}{N}$$

Let us evaluate $X(z)$ at N equally spaced points on unit circle, i.e., at $z = e^{j\frac{2\pi k}{N}}$

$$\therefore X(z) \Big|_{z = e^{j\frac{2\pi k}{N}}} = \sum_{n=0}^{N-1} x(n)z^{-n} \Big|_{z = e^{j\frac{2\pi k}{N}}} = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} \dots\dots(4.1)$$

By the definition of N-point DFT we get,

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}} \dots\dots(4.2)$$

From equations (4.1) and (4.2) we can say that,

$$X(k) = X(z) \Big|_{z = e^{j\frac{2\pi k}{N}}} \dots\dots(4.3)$$

Since the evaluation is performed on unit circle the ROC of $X(z)$ should include unit circle.

5. Analysis of LTI Discrete Time Systems Using DFT

In lecture 10, it is shown that Fourier transform is a useful tool for the analysis of discrete time systems in frequency domain. But the drawback in Fourier transform is that it is a continuous function of ω and so it will not be useful for digital processing of signals and systems. Hence DFT is proposed, therefore the analysis of discrete time systems in frequency domain can be conveniently performed using DFT for digital processing of signals and systems.

Discrete Frequency Spectrum

In general the DFT of a signal gives the discrete frequency spectrum of a signal.

Let $x(n)$ and $X(k)$ be a DFT pair.

Now, $X(k)$ = Discrete frequency spectrum of discrete time signal.

$|X(k)|$ = Magnitude spectrum of discrete time signal.

$\angle X(k)$ = Phase spectrum of discrete time signal.

5. Analysis of LTI Discrete Time Systems Using DFT

In particular, the DFT of impulse response, $h(n)$ of a discrete time system gives discrete frequency response or frequency spectrum of the discrete time system.

Let $h(n)$ and $H(k)$ be a DFT pair.

Now, $H(k)$ = Discrete frequency spectrum of discrete time system.

$|H(k)|$ = Magnitude spectrum of discrete time system.

$\angle H(k)$ = Phase spectrum of discrete time system.

Response of LTI Discrete Time System Using DFT

The response of an LTI discrete time system is given by linear convolution of input and impulse response of the system.

Let, $x(n)$ = Input to an LTI system

$h(n)$ = Impulse response of the LTI system

Now, the response or output of the system $y(n)$ is given by,

$$y(n) = x(n) * h(n) = h(n) * x(n)$$

$$\text{where, } x(n) * h(n) = \sum_{m=-\infty}^{+\infty} x(m) h(n-m) \dots (5.1)$$

5. Analysis of LTI Discrete Time Systems Using DFT

The *DFT supports only circular convolution* and so, the linear convolution of equation (5.1) has to be computed via circular convolution. If $x(n)$ is N_1 -point sequence and $h(n)$ is N_2 -point sequence then linear convolution $x(n)$ and $h(n)$ will generate $y(n)$ of size $N_1 + N_2 - 1$. Therefore in order to perform linear convolution via circular convolution the $x(n)$ and $h(n)$ should be converted to $N_1 + N_2 - 1$ point sequences by appending zeros. Now the circular convolution of $N_1 + N_2 - 1$ point sequences $x(n)$ and $h(n)$ will give same result as that of linear convolution.

Let, $x(n)$ be N_1 -point sequence and $h(n)$ be N_2 -point sequence. Let us convert $x(n)$ and $h(n)$ to $N_1 + N_2 - 1$ point sequences.

Let, $Y(k) = N_1 + N_2 - 1$ point DFT of $y(n)$

$X(k) = N_1 + N_2 - 1$ point DFT of $x(n)$

$H(k) = N_1 + N_2 - 1$ point DFT of $h(n)$ }

5. Analysis of LTI Discrete Time Systems Using DFT

Now by circular convolution theorem of DFT,

$$\mathcal{DFT}\{x(n) \circledast h(n)\} = X(k) H(k)$$

On taking inverse DFT of the above equation we get,

$$x(n) \circledast h(n) = \mathcal{DFT}^{-1}\{X(k) H(k)\}$$

Since, $x(n) \circledast h(n) = y(n)$, the above equation can be written as,

$$y(n) = \mathcal{DFT}^{-1}\{X(k) H(k)\} \dots\dots(5.2)$$

From the equation (5.2) we can say that the output $y(n)$ is given by the inverse DFT of the product of $X(k)$ and $H(k)$.

5. Analysis of LTI Discrete Time Systems Using DFT

Example 1

Compute 4-point DFT and 8-point DFT of three sample sequence given by,

$$\begin{aligned}x(n) &= \frac{1}{3} \quad ; \quad 0 \leq n \leq 2 \\ &= 0 \quad ; \quad \text{else}\end{aligned}$$

Solution

By the definition of N-point DFT, the k^{th} complex coefficient of $X(k)$, for $0 \leq k \leq N - 1$, is given by,

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi kn}{N}}$$

5. Analysis of LTI Discrete Time Systems Using DFT

a) 4-point DFT ($\therefore N = 4$)

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

$$\begin{aligned} X(k) &= \sum_{n=0}^3 x(n) e^{\frac{-j2\pi kn}{4}} = \sum_{n=0}^2 x(n) e^{\frac{-j\pi kn}{2}} = x(0) e^0 + x(1) e^{\frac{-j\pi k}{2}} + x(2) e^{-j\pi k} \\ &= \frac{1}{3} + \frac{1}{3} e^{\frac{-j\pi k}{2}} + \frac{1}{3} e^{-j\pi k} = \frac{1}{3} \left[1 + \cos\frac{\pi k}{2} - j\sin\frac{\pi k}{2} + \cos\pi k - j\sin\pi k \right] \end{aligned}$$

For 4-point DFT, $X(k)$ has to be evaluated for $k = 0, 1, 2, 3$.

$$\begin{aligned} \text{When } k = 0; X(0) &= \frac{1}{3} [1 + \cos 0 - j\sin 0 + \cos 0 - j\sin 0] \\ &= \frac{1}{3} (1 + 1 + 1) = 1 = 1 \angle 0 \end{aligned}$$

$$\begin{aligned} \text{When } k = 1; X(1) &= \frac{1}{3} \left[1 + \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} + \cos\pi - j\sin\pi \right] \\ &= \frac{1}{3} (1 + 0 - j - 1 - j0) = -j\frac{1}{3} = \frac{1}{3} \angle -\pi/2 = 0.333 \angle -0.5\pi \end{aligned}$$

5. Analysis of LTI Discrete Time Systems Using DFT

a) 4-point DFT ($\therefore N = 4$)

$$\begin{aligned}\text{When } k = 2; X(2) &= \frac{1}{3}[1 + \cos \pi - j\sin \pi + \cos 2\pi - j\sin 2\pi] \\ &= \frac{1}{3}(1 - 1 - j0 + 1 - j0) = \frac{1}{3} = 0.333 \angle 0\end{aligned}$$

$$\begin{aligned}\text{When } k = 3; X(3) &= \frac{1}{3}\left[1 + \cos \frac{3\pi}{2} - j\sin \frac{3\pi}{2} + \cos 3\pi - j\sin 3\pi\right] \\ &= \frac{1}{3}(1 + 0 + j - 1 - j0) = j\frac{1}{3} = \frac{1}{3} \angle \pi/2 = 0.333 \angle 0.5\pi\end{aligned}$$

$\therefore$ The 4-point DFT sequence $X(k)$ is given by,

$$X(k) = \{ 1 \angle 0, 0.333 \angle -0.5\pi, 0.333 \angle 0, 0.333 \angle 0.5\pi \}$$

Phase angles
are in radians

$$\therefore \text{Magnitude Function, } |X(k)| = \{ 1, 0.333, 0.333, 0.333 \}$$

$$\text{Phase Function, } \angle X(k) = \{ 0, -0.5\pi, 0, 0.5\pi \}$$

5. Analysis of LTI Discrete Time Systems Using DFT

b) 8-point DFT ($\therefore N = 8$)

$$\begin{aligned} X(k) &= \sum_{n=0}^7 x(n) e^{\frac{-j2\pi kn}{8}} = \sum_{n=0}^2 x(n) e^{\frac{-j\pi kn}{4}} = x(0) e^0 + x(1) e^{\frac{-j\pi k}{4}} + x(2) e^{\frac{-j\pi k}{2}} \quad \boxed{e^{\pm j\theta} = \cos\theta \pm j\sin\theta} \\ &= \frac{1}{3} + \frac{1}{3} e^{\frac{-j\pi k}{4}} + \frac{1}{3} e^{\frac{-j\pi k}{2}} = \frac{1}{3} \left[1 + \cos\frac{\pi k}{4} - j\sin\frac{\pi k}{4} + \cos\frac{\pi k}{2} - j\sin\frac{\pi k}{2} \right] \end{aligned}$$

For 8-point DFT, $X(k)$ has to be evaluated for $k = 0, 1, 2, 3, 4, 5, 6, 7$.

$$\begin{aligned} \text{When } k = 0 ; X(0) &= \frac{1}{3} [1 + \cos 0 - j\sin 0 + \cos 0 - j\sin 0] \\ &= \frac{1}{3} (1 + 1 - j0 + 1 - j0) = 1 = 1 \angle 0 \end{aligned}$$

$$\begin{aligned} \text{When } k = 1 ; X(1) &= \frac{1}{3} \left[1 + \cos\frac{\pi}{4} - j\sin\frac{\pi}{4} + \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} \right] \\ &= 0.333 (1 + 0.707 - j0.707 + 0 - j1) \\ &= 0.568 - j0.568 = 0.803 \angle -0.785 = 0.803 \angle -0.25\pi \end{aligned}$$

5. Analysis of LTI Discrete Time Systems Using DFT

b) 8-point DFT ($\therefore N = 8$)

$$\begin{aligned}\text{When } k = 2; X(2) &= \frac{1}{3} \left[1 + \cos \frac{2\pi}{4} - j \sin \frac{2\pi}{4} + \cos \frac{2\pi}{2} - j \sin \frac{2\pi}{2} \right] \\ &= 0.333 (1 + 0 - j1 - 1 - j0) \\ &= -j0.333 = 0.333 \angle -\pi/2 = 0.333 \angle -0.5\pi\end{aligned}$$

$$\begin{aligned}\text{When } k = 3; X(3) &= \frac{1}{3} \left[1 + \cos \frac{3\pi}{4} - j \sin \frac{3\pi}{4} + \cos \frac{3\pi}{2} - j \sin \frac{3\pi}{2} \right] \\ &= 0.333 (1 - 0.707 - j0.707 + 0 + j1) \\ &= 0.098 + j0.098 = 0.139 \angle -0.785 = 0.139 \angle -0.25\pi\end{aligned}$$

$$\begin{aligned}\text{When } k = 4; X(4) &= \frac{1}{3} \left[1 + \cos \frac{4\pi}{4} - j \sin \frac{4\pi}{4} + \cos \frac{4\pi}{2} - j \sin \frac{4\pi}{2} \right] \\ &= 0.333 (1 - 1 - j0 + 1 - j0) = 0.333 = 0.333 \angle 0\end{aligned}$$

$$\begin{aligned}\text{When } k = 5; X(5) &= \frac{1}{3} \left[1 + \cos \frac{5\pi}{4} - j \sin \frac{5\pi}{4} + \cos \frac{5\pi}{2} - j \sin \frac{5\pi}{2} \right] \\ &= 0.333 (1 - 0.707 + j0.707 + 0 - j1) \\ &= 0.098 - j0.098 = 0.139 \angle -0.785 = 0.139 \angle -0.25\pi\end{aligned}$$

5. Analysis of LTI Discrete Time Systems Using DFT

b) 8-point DFT ($\therefore N = 8$)

$$\begin{aligned}\text{When } k = 6 ; X(6) &= \frac{1}{3} \left[1 + \cos \frac{6\pi}{4} - j \sin \frac{6\pi}{4} + \cos \frac{6\pi}{2} - j \sin \frac{6\pi}{2} \right] \\ &= 0.333 (1 + 0 + j1 - 1 - j0) \\ &= j0.333 = 0.333 \angle \pi / 2 = 0.333 \angle 0.5\pi\end{aligned}$$

$$\begin{aligned}\text{When } k = 7 ; X(7) &= \frac{1}{3} \left[1 + \cos \frac{7\pi}{4} - j \sin \frac{7\pi}{4} + \cos \frac{7\pi}{2} - j \sin \frac{7\pi}{2} \right] \\ &= 0.333 (1 + 0.707 + j0.707 + 0 + j1) \\ &= 0.568 + j0.568 = 0.803 \angle 0.785 = 0.803 \angle 0.25\pi\end{aligned}$$

$\therefore$ The 8-point DFT sequence $X(k)$ is given by,

$$X(k) = \{1 \angle 0, 0.803 \angle -0.25\pi, 0.333 \angle -0.5\pi, 0.139 \angle -0.25\pi, 0.333 \angle 0, 0.139 \angle 0.25\pi, 0.333 \angle 0.5\pi, 0.803 \angle 0.25\pi\}$$

$$\therefore \text{ Magnitude Function, } |X(k)| = \{1, 0.803, 0.333, 0.139, 0.333, 0.139, 0.333, 0.803\}$$

$$\text{Phase Function, } \angle X(k) = \{0, -0.25\pi, -0.5\pi, -0.25\pi, 0, 0.25\pi, 0.5\pi, 0.25\pi\}$$

5. Analysis of LTI Discrete Time Systems Using DFT

Example 2: (Home Work)

Compute the DFT of the sequence, $x(n) = \{0, 1, 2, 3\}$. Sketch the magnitude and phase spectrum.

5. Analysis of LTI Discrete Time Systems Using DFT

Example 3

Compute circular convolution of the following two sequences using DFT.

$$x_1(n) = \{ \underset{\uparrow}{2}, 1, 2, 1 \} \text{ and } x_2(n) = \{ \underset{\uparrow}{1}, 2, 3, 4 \}$$

Solution

Given that, $x_1(n) = \{ 2, 1, 2, 1 \}$. The 4-point DFT of $x_1(n)$ is,

$$\begin{aligned} \mathcal{DFT}\{x_1(n)\} &= X_1(k) = \sum_{n=0}^3 x_1(n) e^{\frac{-j2\pi nk}{4}} ; k = 0, 1, 2, 3 \\ &= x_1(0) e^0 + x_1(1) e^{\frac{-j2\pi k}{4}} + x_1(2) e^{\frac{-j2\pi 2k}{4}} + x_1(3) e^{\frac{-j2\pi 3k}{4}} = 2 + e^{\frac{-j\pi k}{2}} + 2 e^{-j\pi k} + e^{\frac{-j3\pi k}{2}} \end{aligned}$$

$$\text{When } k = 0 ; X_1(0) = 2 + 1 + 2 + 1 = 6$$

$$\text{When } k = 1 ; X_1(1) = 2 + e^{\frac{-j\pi}{2}} + 2 e^{-j\pi} + e^{\frac{-j3\pi}{2}} = 2 - j - 2 + j = 0$$

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

5. Analysis of LTI Discrete Time Systems Using DFT

$$\text{When } k = 2 ; X_1(2) = 2 + e^{-j\pi} + 2e^{-j2\pi} + e^{-j3\pi} = 2 - 1 + 2 - 1 = 2$$

$$\text{When } k = 3 ; X_1(3) = 2 + e^{\frac{-j3\pi}{2}} + 2e^{-j3\pi} + e^{\frac{-j9\pi}{2}} = 2 + j - 2 - j = 0$$

Given that, $x_2(n) = \{ 1, 2, 3, 4 \}$. The 4-point DFT of $x_2(n)$ is,

$$\begin{aligned} \mathcal{DFT}\{x_2(n)\} &= X_2(k) = \sum_{n=0}^3 x_2(n) e^{\frac{-j2\pi nk}{4}} ; k = 0, 1, 2, 3 \\ &= x_2(0) e^0 + x_2(1) e^{\frac{-j\pi k}{2}} + x_2(2) e^{-j\pi k} + x_2(3) e^{\frac{-j3\pi k}{2}} \\ &= 1 + 2e^{\frac{-j\pi k}{2}} + 3e^{-j\pi k} + 4e^{\frac{-j3\pi k}{2}} \end{aligned}$$

$$\text{When } k = 0 ; X_2(0) = 1 + 2 + 3 + 4 = 10$$

$$\text{When } k = 1 ; X_2(1) = 1 + 2e^{\frac{-j\pi}{2}} + 3e^{-j\pi} + 4e^{\frac{-j3\pi}{2}} = 1 - 2j - 3 + 4j = -2 + j2$$

$$\text{When } k = 2 ; X_2(2) = 1 + 2e^{-j\pi} + 3e^{-j2\pi} + 4e^{-j3\pi} = 1 - 2 + 3 - 4 = -2$$

5. Analysis of LTI Discrete Time Systems Using DFT

$$\text{When } k = 3; X_2(3) = 1 + 2e^{\frac{-j3\pi}{2}} + 3e^{-j3\pi} + 4e^{\frac{-j9\pi}{2}} = 1 + 2j - 3 - 4j = -2 - j2$$

$$X_1(k) = \begin{cases} 6 & ; k = 0 \\ 0 & ; k = 1 \\ 2 & ; k = 2 \\ 0 & ; k = 3 \end{cases}$$

$$X_2(k) = \begin{cases} 10 & ; k = 0 \\ -2 + j2 & ; k = 1 \\ -2 & ; k = 2 \\ -2 - j2 & ; k = 3 \end{cases}$$

Let $X_3(k)$ be the product of $X_1(k)$ and $X_2(k)$.

$$\therefore X_3(k) = X_1(k) X_2(k)$$

$$\text{When } k = 0; \quad X_3(0) = X_1(0) \times X_2(0) = 6 \times 10 = 60$$

$$\text{When } k = 1; \quad X_3(1) = X_1(1) \times X_2(1) = 0 \times (-2 + j2) = 0$$

$$\text{When } k = 2; \quad X_3(2) = X_1(2) \times X_2(2) = 2 \times (-2) = -4$$

$$\text{When } k = 3; \quad X_3(3) = X_1(3) \times X_2(3) = 0 \times (-2 - j2) = 0$$

$$\therefore X_3(k) = \{60, 0, -4, 0\}$$

5. Analysis of LTI Discrete Time Systems Using DFT

By circular convolution theorem of DFT we get,

$$\mathcal{DFT}\{x_1(n) \circledast x_2(n)\} = X_1(k) X_2(k) \quad \Rightarrow \quad x_1(n) \circledast x_2(n) = \mathcal{DFT}^{-1} \{ X_1(k) X_2(k) \} = \mathcal{DFT}^{-1} \{ X_3(k) \}$$

Let $x_3(n)$ be the sequence obtained by taking inverse DFT of $X_3(k)$.

$$\mathcal{DFT}^{-1} \{ X_3(k) \} = x_3(n) = \frac{1}{N} \sum_{k=0}^{N-1} X_3(k) e^{\frac{j2\pi nk}{4}} \quad ; \quad n = 0, 1, 2, 3$$

$$= \frac{1}{4} [X_3(0) e^0 + X_3(2) e^{j\pi n}] = \frac{1}{4} [60 - 4 e^{j\pi n}]$$

$$\text{When } n = 0; \quad x_3(0) = \frac{1}{4} (60 - 4) = \frac{56}{4} = 14$$

$$\text{When } n = 1; \quad x_3(1) = \frac{1}{4} (60 - 4e^{j\pi}) = \frac{64}{4} = 16$$

$$\text{When } n = 2; \quad x_3(2) = \frac{1}{4} (60 - 4e^{j2\pi}) = \frac{56}{4} = 14$$

$$\text{When } n = 3; \quad x_3(3) = \frac{1}{4} (60 - 4e^{j3\pi}) = \frac{64}{4} = 16$$

$$\therefore x_1(n) \circledast x_2(n) = x_3(n) = \{14, 16, 14, 16\}$$

5. Analysis of LTI Discrete Time Systems Using DFT

Example 4: (Home Work)

Compute circular convolution of the following two sequences using DFT.

$$x(n) = \{1, 0.5\} \quad \text{and} \quad h(n) = \{0.5, 1\}$$

$\uparrow$ $\uparrow$

6. Fast Fourier Transform (FFT)

- The **Fast Fourier Transform (FFT)** is a method (or algorithm) for computing the discrete Fourier transform (DFT) with reduced number of calculations. The computational efficiency is achieved if we adopt a divide and conquer approach. This approach is based on the decomposition of an N -point DFT into successively smaller DFTs. This basic approach leads to a family of an efficient computational algorithms known collectively as FFT algorithms.
- **Radix- r FFT**- In an N -point sequence, if N can be expressed as $N = r^m$, then the sequence can be decimated into r -point sequences. For each r -point sequence, r -point DFT can be computed. From the results of r -point DFT, the r^2 -point DFTs are computed. From the results of r^2 -point DFTs, the r^3 -point DFTs are computed and so on, until we get r^m point DFT. This FFT algorithm is called radix- r FFT. In computing N -point DFT by this method the number of stages of computation will be m times.

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6. Fast Fourier Transform (FFT)

- **Radix-2 FFT-** For radix-2 FFT, the value of N should be such that, $N = 2^m$, so that the N -point sequence is decimated into 2-point sequences and the 2-point DFT for each decimated sequence is computed. From the results of 2-point DFTs, the 4-point DFTs can be computed. From the results of 4-point DFTs, the 8-point DFTs can be computed and so on, until we get N -point DFT.

6. Fast Fourier Transform (FFT)

Number of Calculations in N-point DFT

Let, $X(k)$ be N-point DFT of an L-point discrete time sequence $x(n)$, where $N \geq L$. Now, the N-point DFT is a sequence consisting of N-complex numbers. Each complex number of the sequence is calculated using the following equation, (equation 2.2).

$$\begin{aligned} X(k) &= \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi kn}{N}} ; \text{ for } k = 0, 1, 2, \dots, N-1 \\ &= x(0) e^0 + x(1) e^{\frac{-j2\pi k}{N}} + x(2) e^{\frac{-j4\pi k}{N}} + x(3) e^{\frac{-j6\pi k}{N}} + \dots + x(N-1) e^{\frac{-j2(N-1)\pi k}{N}} \end{aligned}$$

The computation of above equation for one value of k involves N number of complex multiplications and $N-1$ number of complex additions. Therefore, the computation of all the N values of the sequence $X(k)$ involves $N \times N = N^2$ complex multiplications and $N \times (N-1)$ complex additions. Hence, in direct computation of N-point DFT, the total number of complex additions are $N(N-1)$ and total number of complex multiplications are N^2 . *(Refer to example 3 to understand this concept)*

6. Fast Fourier Transform (FFT)

- **Number of Calculations in Radix-2 FFT**

* In Radix-2 FFT, $N = 2^m$, there will be 'm' stages of computations where $m = \log_2(N)$

* Each stage = $\frac{N}{2}$ butterflies

where 1 butterfly = 1 complex multiplication + 2 complex additions.

* No. of computations in each stage are:

> No. of complex additions = $2 \times \frac{N}{2} = N$

> No. of complex multiplications = $1 \times \frac{N}{2} = \frac{N}{2}$

* Therefore, the total number of computations for m-stages will be:

6. Fast Fourier Transform (FFT)

- **Number of Calculations in Radix-2 FFT**

$N \times m = N \log_2(N)$ complex additions

$\left(\frac{N}{2}\right) \times m = \left(\frac{N}{2}\right) \log_2(N)$ complex multiplications

* Hence in radix-2 FFT, the total number of complex additions are reduced to $N \log_2(N)$ and the total number of complex multiplications are reduced to $\left(\frac{N}{2}\right) \log_2 N$.

6. Fast Fourier Transform (FFT)

The table 2 presents a comparison of the number of complex multiplications and additions in radix-2 FFT and in direct computation of DFT. From the table it can be observed that for large values of N , the percentage reduction in calculations is also very large.

$$\log_2 2^m = m$$

$$\log_y x = \frac{\log_{10} x}{\log_{10} y}$$

Table 2 : Comparison of Number of Computation in Direct DFT and FFT

Number of points N	Direct Computation		Radix-2 FFT	
	Complex additions $N(N-1)$	Complex Multiplications N^2	Complex additions $N \log_2 N$	Complex Multiplications $(N/2) \log_2 N$
4 ($= 2^2$)	12	16	$4 \times \log_2 2^2 = 8$	$\frac{4}{2} \times \log_2 2^2 = 4$
8 ($= 2^3$)	56	64	$8 \times \log_2 2^3 = 24$	$\frac{8}{2} \times \log_2 2^3 = 12$
16 ($= 2^4$)	240	256	$16 \times \log_2 2^4 = 64$	$\frac{16}{2} \times \log_2 2^4 = 32$
32 ($= 2^5$)	992	1,024	$32 \times \log_2 2^5 = 160$	$\frac{32}{2} \times \log_2 2^5 = 80$
64 ($= 2^6$)	4,032	4,096	$64 \times \log_2 2^6 = 384$	$\frac{64}{2} \times \log_2 2^6 = 192$
128 ($= 2^7$)	16,256	16,384	$128 \times \log_2 2^7 = 896$	$\frac{128}{2} \times \log_2 2^7 = 448$

6. Fast Fourier Transform (FFT)

Phase or Twiddle Factor

By the definition of DFT, the N-point DFT is given by,

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi nk}{N}} ; \text{ for } k = 0, 1, 2, \dots, N-1 \dots (6.1)$$

To simplify the notation it is desirable to define the complex valued phase factor W_N (also called as twiddle factor)

$$W_N = e^{\frac{-j2\pi}{N}}$$

Here, W represents a complex number $1 \angle -2\pi$. Hence the phase or argument of W is -2π . Therefore, when a number is multiplied by W , only its phase changes by -2π but magnitude remains same.

$$\therefore W = e^{-j2\pi}$$

$$W_1 = e^{\frac{-j2\pi}{1}}$$

6. Fast Fourier Transform (FFT)

The phase value -2π of W can be multiplied by any integer and it is represented as prefix in W . For example multiplying -2π by k can be represented as W^k .

$$\therefore e^{-j2\pi \times k} \Rightarrow W^k$$

The phase value -2π of W can be divided by any integer and it is represented as suffix in W . For example dividing -2π by N can be represented as W_N .

$$\begin{aligned}\therefore e^{-j2\pi \div N} &= e^{-j2\pi \times \frac{1}{N}} \Rightarrow W_N \\ \therefore e^{\frac{-j2\pi nk}{N}} &= \left(e^{-j2\pi}\right)^{\frac{nk}{N}} = W_N^{nk} \dots\dots(6.2)\end{aligned}$$

$$\left(e^{\frac{-j2\pi}{N}}\right)^{nk} = W_N^{nk}$$

Using equation (6.2) the equation (6.1) can be written as,

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad ; \text{ for } k = 0, 1, 2, \dots, N-1 \dots\dots(6.3)$$

The equation (6.3) is the definition of N -point DFT using phase factor, and this equation is popularly used in FFT.

7. Decimation in Time (DIT) Radix-2 FFT

- The N-point DFT of a sequence $x(n)$ converts the time domain N-point sequence $x(n)$ to a frequency domain N-point sequence $X(k)$. In **Decimation In Time (DIT)** algorithm, the time domain sequence $x(n)$ is decimated and smaller point DFTs are performed. The results of smaller point DFTs are combined to get the result of N-point DFT.
- In DIT radix-2 FFT, the time domain sequence is decimated into 2-point sequences. For each two point sequence, the two point DFT is computed. The results of 2-point DFTs are used to compute 4-point DFTs. The results of 4-point DFTs are used to compute 8-point DFTs. This process is continued until we get N-point DFT.

7. Decimation in Time (DIT) Radix-2 FFT

* Let $x(n)$ be N -sample sequence. We can decimate $x(n)$ into two sequences of $N/2$ samples. Let the two sequences be $f_1(n)$ and $f_2(n)$. Let $f_1(n)$ consists of even numbered samples of $x(n)$ and $f_2(n)$ consists of odd numbered samples of $x(n)$.

$$\therefore f_1(n) = x(2n) : \text{for } n=0, 1, 2, \dots, \frac{N}{2}-1.$$

$$\therefore f_2(n) = x(2n+1) : \text{for } n=0, 1, 2, \dots, \frac{N}{2}-1$$

* Let $X(k)$ = N -point DFT of $x(n)$

$F_1(k)$ = $N/2$ point DFT of $f_1(n)$

$F_2(k)$ = $N/2$ point DFT of $f_2(n)$

7. Decimation in Time (DIT) Radix-2 FFT

* By definition of DFT of $x_1(n)$ and $x_2(n)$ are given by:

$$e^{-j\frac{2\pi nk}{N}} = W_N^{nk}$$

$$F_1(k) = \sum_{n=0}^{N-1} x_1(n) W_N^{kn}$$

$$= \sum_{n=0}^{N-1} x(2n) W_N^{k(2n)}$$

$$= \sum_{n=0}^{N-1} x(2n) W_N^{2kn}$$

Note: $W_N^2 = e^{-j\frac{2\pi}{N}(2)} = e^{-j\frac{2\pi}{N/2}} \rightarrow W_{N/2} = W_{\frac{N}{2}}$

$$F_1(k) = \sum_{n=0}^{N-1} x(2n) W_{\frac{N}{2}}^{kn}$$

7. Decimation in Time (DIT) Radix-2 FFT

Similarly,

$$F_2(k) = \sum_{n=0}^{\frac{N}{2}-1} \frac{x}{2}(n) W_N^{kn}$$

$$F_2(k) = \sum_{n=0}^{\frac{N}{2}-1} x(2n+1) W_N^{k(2n+1)}$$

$$\begin{aligned} F_2(k) &= \sum_{n=0}^{\frac{N}{2}-1} x(2n+1) W_N^{2kn} \cdot W_N^k \\ &= W_N^k \sum_{n=0}^{\frac{N}{2}-1} x(2n+1) \cdot W_{\frac{N}{2}}^{kn} \end{aligned}$$

* Now N -point DFT $X(k)$ in terms of $\frac{N}{2}$ point DFTs $F_1(k)$ and $F_2(k)$ is given by:

$$\boxed{X(k) = F_1(k) + W_N^k F_2(k)}$$

7. Decimation in Time (DIT) Radix-2 FFT

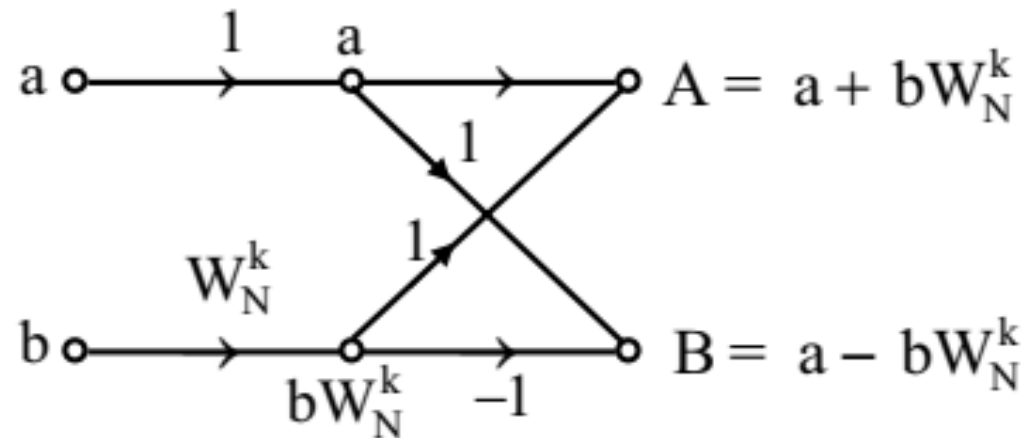
8-Point DFT Using Radix-2 DIT FFT

- The computation of 8-point DFT using radix-2 FFT, involves three stages of computation. Here $N = 8 = 2^3$, therefore $r = 2$ and $m = 3$. The given 8-point sequence is decimated to 2-point sequences. For each 2-point sequence, the 2-point DFT is computed. From the result of 2-point DFT, the 4-point DFT can be computed. From the result of 4-point DFT, the 8-point DFT can be computed.

7. Decimation in Time (DIT) Radix-2 FFT

Basic butterfly or flow graph of DIT radix-2 FFT.

- The signal flow graph is also called butterfly diagram since it resembles a butterfly.
1. In each computation two complex numbers "a" and "b" are considered.
 2. The complex number "b" is multiplied by a phase factor " W_N^k ".
 3. The product " bW_N^k " is added to complex number "a" to form new complex number "A".
 4. The product " bW_N^k " is subtracted from complex number "a" to form new complex number "B".



7. Decimation in Time (DIT) Radix-2 FFT

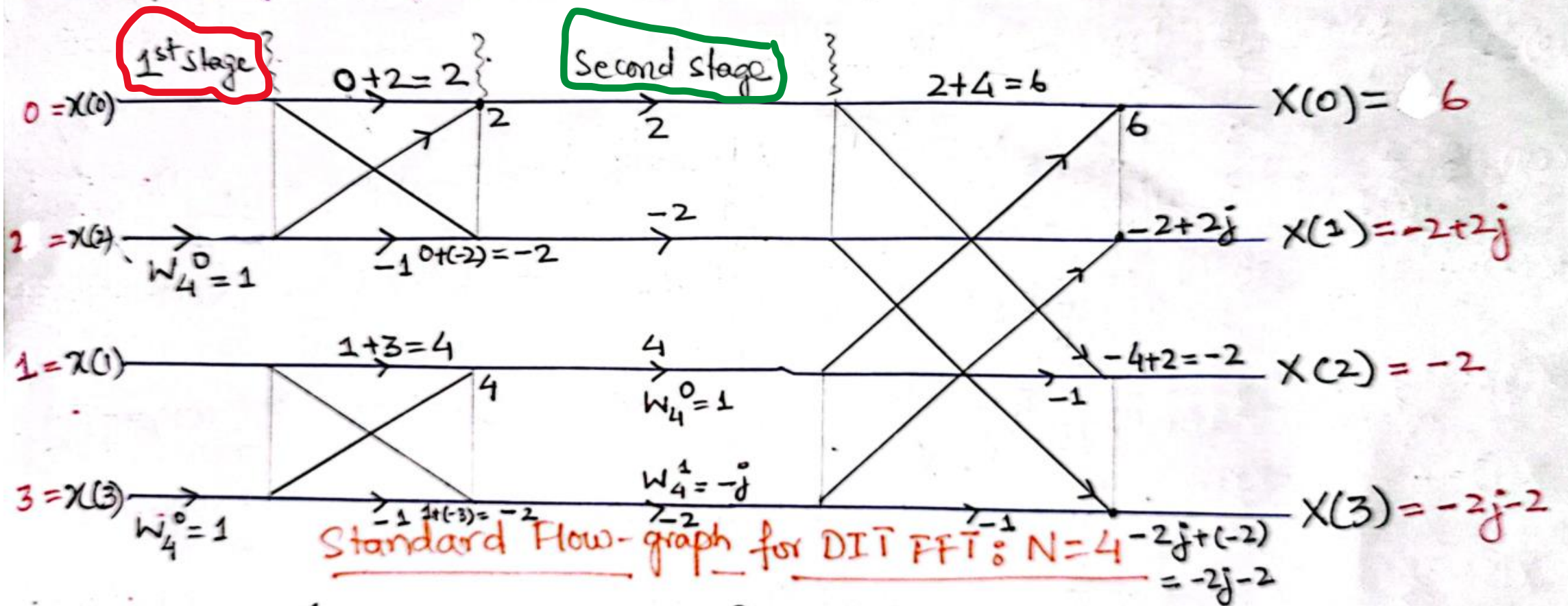
Example 5:

A 4-point sequence is given by $x(n) = \{0, 1, 2, 3, 4\}$. Compute 4-point DFT of $x(n)$ by radix-2-DIT-FFT.

7. Decimation in Time (DIT) Radix-2 FFT

Solution $N=4$ ($N=2^m=2^2=4$; no. of stages = $m=2$)

$N=4$ $\begin{cases} \text{Even} (0, 2) \\ \text{Odd} (1, 3) \end{cases}$



$$X(k) = \{6, -2+2j, -2, -2j-2\}$$

7. Decimation in Time (DIT) Radix-2 FFT

4-point DFT using DIT FFT Phase/Twiddle Factors

$$N = 4; 2^m = 2^2 = 4; 2 \text{ stages.}$$

1st Stage

$$W_2^0 = 1$$

2nd Stage

$$W_4^0 = 1$$

$$W_4^{-1} = e^{-j2\pi \times \frac{1}{4}}$$

$$= e^{-j\frac{\pi}{2}}$$

$$= \cos\left(-\frac{\pi}{2}\right) + j\sin\left(-\frac{\pi}{2}\right)$$

$$W_4^{-1} = -j$$

$$W_N^k = \left(e^{-j\frac{2\pi k}{N}}\right) = e^{-j2\pi \times \frac{k}{N}}$$

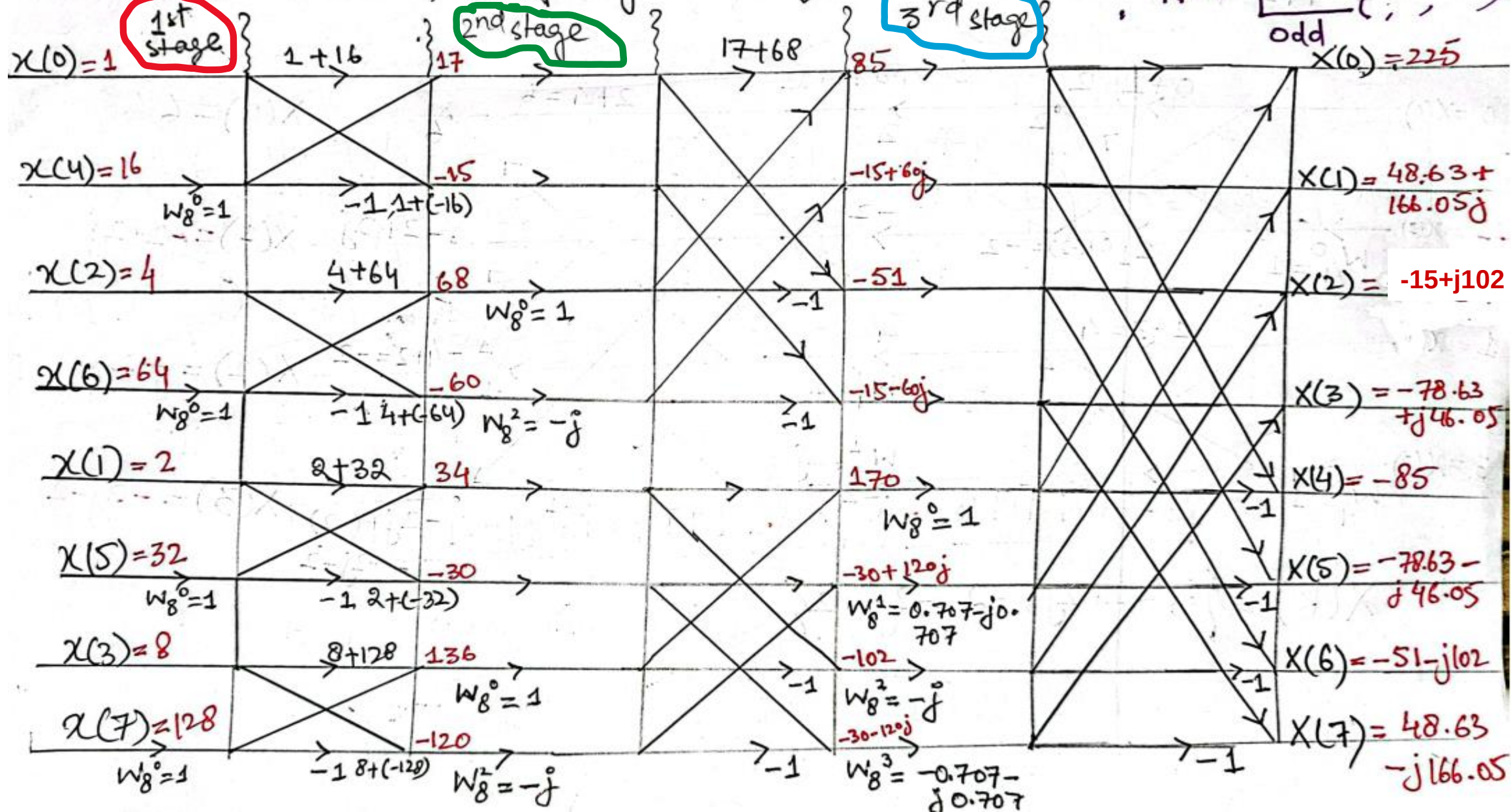
$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

7. Decimation in Time (DIT) Radix-2 FFT

Example 6: An 8-point sequence is given by $x(n) = \{1, 2, 4, 8, 16, 32, 64, 128\}$. Compute 8-point DFT of $x(n)$ by radix-2-DIT-FFT.

$\therefore N=8$ ($N=2^m=2^3$; no. of stages $=m=3$)

$N=8$ - Even (0, 4, 2, 6)
Odd (1, 5, 3, 7)



7. Decimation in Time (DIT) Radix-2 FFT

8-point DFT using DIT FFT Phase/Twiddle Factors

$N=8$; $2^m=2^3=8$; 3 stages ..

$$W_N^k = e^{-j\frac{2\pi k}{N}} = e^{-j2\pi \times \frac{k}{N}}$$

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$

1st stage

$$W_2^0 = 1$$

2nd stage

$$W_4^0 = e^{-j2\pi \times \frac{0}{4}}$$

$$W_4^0 = 1$$

$$W_4^1 = e^{-j2\pi \times \frac{1}{4}}$$

$$= e^{-j\frac{\pi}{2}}$$

$$= \cos(-\frac{\pi}{2}) + j\sin(-\frac{\pi}{2})$$

$$W_4^1 = -j$$

3rd stage

$$W_8^0 = 1$$

$$W_8^1 = e^{-j2\pi \times \frac{1}{8}}$$

$$= e^{-j\frac{\pi}{4}}$$

$$= \cos(-\frac{\pi}{4}) + j\sin(-\frac{\pi}{4})$$

$$W_8^1 = 0.707 - j0.707$$

Similarly

$$W_8^2 = -j$$

$$W_8^3 = -0.707 - j0.707$$

7. Decimation in Time (DIT) Radix-2 FFT

Example 7: (Home Work)

An 8-point sequence is given by $x(n) = \{2, 2, 2, 2, 1, 1, 1, 1\}$. Compute 8-point DFT of $x(n)$ by radix-2-DIT-FFT. Also sketch the magnitude and phase spectrum.

8. Computation of Inverse DFT Using FFT

Let $x(n)$ and $X(k)$ be N -point DFT pair.

Now by the definition of inverse DFT,

$$\begin{aligned} x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{j2\pi nk}{N}} ; \quad \text{for } n = 0, 1, 2, \dots, N-1 \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) \left(e^{\frac{-j2\pi nk}{N}} \right)^* = \frac{1}{N} \sum_{k=0}^{N-1} X(k) (W_N^{nk})^* = \frac{1}{N} \left[\sum_{k=0}^{N-1} X(k) (W_N^{nk})^* \right] \dots (1) \end{aligned}$$

(Refer to Slide no 40)

In equation (1), the expression inside the bracket is similar to that of DFT computation of a sequence, with following differences.

1. The summation index is k instead of n .
2. The input sequence is $X(k)$ instead of $x(n)$.
3. The phase factors are conjugate of the phase factor used for DFT.

the N -point **DFT** of $x(n)$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi kn}{N}} ; \quad \text{for } k = 0, 1, 2, \dots, N-1$$

8. Computation of Inverse DFT Using FFT

Hence, in order to compute inverse DFT of $X(k)$, the FFT algorithm can be used by taking the conjugate of phase factors. Also from equation (1) it is observed that the output of FFT computation should be divided by N to get $x(n)$.

The following procedure can be followed to compute inverse DFT using FFT algorithm.

1. Take N -point frequency domain sequence $X(k)$ as input sequence.
2. Compute FFT by using conjugate of phase factors.
3. Divide the output sequence obtained in FFT computation by N , to get the sequence $x(n)$.

Thus a single FFT algorithm can be used for evaluation of both DFT and inverse DFT.

8. Computation of Inverse DFT Using FFT

Example 8: (Home Work)

In an LTI system the input $x(n) = \{1, 1, 1\}$ and the impulse response $h(n) = \{-1, -1\}$. Determine the response of the LTI system by radix-2 DIT FFT.

Hints:

The various steps in computing $y(n)$ are,

Step - 1 : Determine $X(k)$ using radix-2 DIT algorithm.

Step - 2 : Determine $H(k)$ using radix-2 DIT algorithm.

Step - 3 : Determine the product $X(k)H(k)$.

Step - 4 : Take inverse DFT of the product $X(k)H(k)$ using radix-2 DIT algorithm.

Lengths of $x(n)$, $h(n)$ and $y(n)$ must be same.

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*